

AutoCAD Electrical 2025: AutoLISP Reference

Section A - Schematic Components

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Description

Insert Component, Break Wire, and Auto Tag Generation.

Topics

Name	Description
c:ace_iec_tag_freshen1	Trigger reformat of TAG1/TAG2/TAGSTRIP/P_TAG1 attribute text
c:ace_ins_parametric_connector	Insert parametrically generated multi-pin schematic connector symbol.
c:ace_lastent_list_from	Get list of recent entities inserted into the active drawing. Return entities inserted into the active drawing database after "from_en inserted.
c:ace_lastent_mark	Find the last entity or subentity in the active drawing. Use with (c:ace_lastent_list_from...)
c:ace_multipole	Insert multipole symbol. The spacing between poles is determine intersection with underlying wires or, if none found, by the drawi default ladder rung spacing (carried on attribute RUNGDIST of th drawing's WD_M block insert which then becomes accessible as t 11th element (index 0 based) of the drawing's GBL_wd_m configuration global variable).
c:ace_pnl2sch_lookup	Find schematic component symbol name. For a given panel footp symbol, attempt to find the equivalent schematic component syn name. This data is returned by a query on the schematic_lookup file.
c:ace_projwide_util_dbx	Project-wide update/retag function.
c:ace_retag_project	Re-tag schematic components on all drawings in "dlst" or in entir active project (if "dlst" is nil). This is done in a non-scripting moc
c:wd_calc_comptag	Calculate tag-ID for parent schematic component.
c:wd_delsym_main	Erase connected component "ben" and try to reconnect all broke wires and reconcile duplicate wire numbers.
c:wd_get_pinlist	Query pin list database and return pin list info.
c:wd_ins_circ2	User hook into the "INSERT CIRC" command to insert a collector prewired components. AcadE automatically retags the inserted components based upon circuit line references or next available sequential.
c:wd_insert_elect_block	Generic INSERT BLOCK command. Wires do NOT break under the block.
c:wd_insum	User hook into the INS COMPONENT command. command-line

	(if required), component tag automatically generates based on inserted line ref or next available sequential.
c:wd_insym_dlg	User hook into the INS COMPONENT command, icon menu driver. Same as wd_insym but command triggers icon menu display for component selection, user selects insertion point. The icon menu pops up is the schematic icon menu that is in memory or is the default schematic icon menu (if the menu isn't already in memory). Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.
c:wd_insym_f	User hook into the INS COMPONENT command, prompt to select "insert fence" component. User prompted to select "insert fence" component from icon menu. Then user prompted to draw fence. Inserts selected component at intersections between fence and underlying wires.
c:wd_insym_f_same	User hook into the INS COMPONENT command, user prompted to pick on a "just like" component. Same as wd_insym_f but user prompted to pick on a "just like" component. Then user prompted to draw fence. AcadE inserts selected component at intersections between fence and underlying wires.
c:wd_insym_qo2menu	Display a submenu page in the current icon menu. Jump to and display the "mnum" submenu page in the current icon menu file. A pick on this menu will feed the selected button's value to the INS COMPONENT command.
c:wd_insym_qomenu	Jump to and display the "mnum" submenu page in the current icon menu file. A pick on this menu will feed the selected button's value to the INS COMPONENT command.
c:wd_insym_same	User hook into the INS COMPONENT command, just-like component insert prompted. Same as wd_insym but user prompted to select "just like" component (schematic or panel footprint). Then user is prompted for insertion point. INS/EDIT dialog box appears after insertion. The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon menu (if the menu isn't already in memory). Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.
c:wd_insym2	User hook into the INS COMPONENT command. All data is supplied on the command line with optional EDIT dialog box popping up after symbol inserts. Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.
c:wd_insymn	User hook into the INS COMPONENT command, name via command line, insert prompted. Same as wd_insym but component name supplied on command line, user prompted to select insertion point. INS/EDIT dialog box displays after insert. The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon menu (if the menu isn't already in memory). Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.

with any underlying wire. The Insert/Edit dialog box appears after insert. The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon menu (if the menu isn't already in memory. Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically... [more](#)

c:wd_insymn_rot	User hook into the INS COMPONENT command, name via command line, insert prompted, rotation prompted. Component name supplied on command line, user prompted to select insertion point, but if device cannot find a wire/line connection to tie to, it then prompts user for an insertion ROTATION value. The INS/EDIT dialog box appears after insert. The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon menu (if the menu isn't already in memory. Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required),... more
c:wd_loadmenu	Activate AcadE icon menu. Make AcadE icon menu file "fnam" the active icon menu (for INSERT COMPONENT using c:wd_insym_dlg, c:wd_insym_f, and others).
c:wd_menu_load_schem	Loads the default schematic icon menu .dat file is loaded and "in memory" so that it will display for a subsequent call related to INS COMPONENT. The menu that is loaded is determined in this order - an override menu name is defined in global GBL_wd_sch_menu_override. 2nd - defined for the active project (right-click on project's "Symbol Libraries"). 3rd - defaults to menu file "ace_jic_menu.dat". This routine can be used in ".cui" file call "wd_insym_gomenu mnum" so that AcadE will display schematic menu pages and not menu pages from the last icon menu file used (if... more
c:wd_movesym2	User hook into the MOVE COMPONENT command, all data supplied on the command line. Moved component automatically aligns with a breaks underlying wire(s), new wire number generates on new wire segment (if required), component tag can automatically generate based on inserted line ref or next available sequential. Wire "head" where component moved from.
c:wd_pinlist_attach	Populate parent component with PINLIST info based upon its mfg combo carried on the symbol. If a match is found in the _PINLIST table of the catalog lookup file then that data is written to the symbol PINLIST attrib (or xdata).
c:wd_retag_update_noprompts	Re-tag schematic component(s) on current dwg only and optionally updates related child components across active project.
c:wd_xref_doit	Run project-wide parent/child cross-reference text update. Note: drawing or drawings to be processed must be part of the active project.

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Description

Trigger reformat of TAG1/TAG2/TAGSTRIP/P_TAG1 attribute text.

AutoLISP

```
(c:ace_iec_tag_freshen1 param_lst)
```

Parameters

Parameters	Description
param_lst	(list ben options) where: "ben" = component or panel footprint entity to update. "options" = 4's bit set = force INST/LOC attribs to be VISIBLE, 8's bit set = force INST/LOC attribs to be INVISIBLE.

Returns

1 if something changed, nil otherwise.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_ins_parametric_connector
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Description

Insert parametrically generated multi-pin schematic connector symbol.

AutoLISP

```
(c:ace_ins_parametric_connector xy scl param_lst)
```

xy	Insertion point. nil = prompt for insertion point.
scl	nil = default scale (follows global GBL_wd_scl_plcbl)
param_lst	List of parameters nth 0 : connector type (1=plug/recept, 2= plug/recept (2 blocks), 4=plug only, 8=receptacle only) nth 1 : show pin numbers - plug/jack: 2 = both, 4 = side, 8 = recpt side, 16 = neither side, single plug or single jack insert: 0=hide, 1=show nth 2 : 1 = vertical insert, 2 = horizontal nth 3 : 1 = plug side on right or top, 2 = left or bottom nth 4 : 1 = add divider line if plug/jack insert, 0 = no divider line nth 5 : 1 = wire number changes through plug/jack symbol nth 6 : 1 = wire number changes through plug/jack symbol nth 7 : 1 = fixed pin spacing, 2 = stretch to match wire crossings nth 8 : 1 = insert all, 2 = prompt spacers/breaks nth 9 : pin spacing (ex: 0.75) nth 10 : receptacle side width nth 11 : plug side width nth 12 : top extra height nth 13 : bottom extra height nth 14 : plug corner radius value nth 15 : pin count nth 16 : starting pin number text string (or comma-delimited string of specific pin assignments, like "A,E") nth 17 : 1 = start insert as a "child" connector, 0 = normal nth 18 : 1 = start with "break" line, 0 = normal

Returns

if fail, otherwise list of entity names of connector or connector pieces that inserted

Example

```
(if (not GBL_wd_m)(c:ace_gbl_wd_m 1))  
(c:ace_ins_parametric_connector (list 5.0 10.5 0.0) nil (list 1 4 1 2 1 0 0 1 1 0.75 0.25 0.25 0.25 0.25 0.125 3 "1"))
```

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Description

Get list of recent entities inserted into the active drawing. Return list of entities inserted into the active drawing dat after "from_ent" was inserted.

AutoLISP

```
(c:ace_lastent_list_from from_ent options)
```

Parameters

Parameters	Description
from_ent	The return value from (c:ace_lastent_mark) or nil=s beginning of database.
options	1's bit set = return all entities, even subentities. 1's bit clear = return only block INSERT entities. 2's bit set = return as list of "handles" (clear = retur of entity names).

Returns

List of handles or entity names (nil=none found).

Example

Insert a 3-pole circuit breaker, mark each block's RATING1 attribute with a value of "30 AMPS".

```
(setq last_one (c:ace_lastent_mark)) ; remember last entity in database before inserting 3-pole  
breaker  
(c:ace_multipole "HCB1" 3 (list 5.5 6.6 0.0) 1)  
(setq ent_list (c:ace_lastent_list_from last_one 0)) ; get list of block inserts that were adde  
(foreach ent ent_list (c:wd_modattrval ent "RATING1" "30 AMPS" nil))
```

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Find the last entity or subentity in the active drawing. Use with (c:ace_lastent_list_from...)

AutoLISP

```
(c:ace_lastent_mark)
```

Returns

Last entity name (or nil if drawing is empty).

Example

```
(setq last_one (c:ace_lastent_mark))
```

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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_multipole

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Description

Insert multipole symbol. The spacing between poles is determined by intersection with underlying wires or, if none by the drawing's default ladder rung spacing (carried on attribute RUNGDIST of the drawing's WD_M block insert wl then becomes accessible as the 11th element (index 0 based) of the drawing's GBL_wd_m configuration global vari

AutoLISP

```
(c:ace_multipole symname polecnt xy options)
```

Parameters

Parameters	Description
symname	Symbol to insert.
polecnt	Total pole count (nil, 1, 2 =2 pole). 2 pole is minimu

component inserted)
 bit 2 set means force default to start with "H" symbol orientation (horiz wire insert)
 bit 4 set means force default to start with "V" symbol orientation (vertical wire insert)
 bit 2+4 blank = no forced default
 bit 8 set means force build vertical wire insertion to LEFT or horizontal wire insertion UP
 bit 16 set means force build vertical wire insertion to RIGHT or horizontal wire insertion DOWN
 bit 8+16 blank means prompt user for direction.
 bit 64 suppress any prompt for user pick on child wire not found)
 bit 128 set means that all LINES are viewed as valid and can be connected to, even if not on a wire layer by CIRC BLDR utility)

Example 1

Insert 2-pole circuit breaker at 5.5,6.6.

```
(c:ace_multipole "HCB1" 2 (list 5.5 6.6 0.0) 1)
```

Example 2

Insert 3-pole disconnect switch at 5.5,6.6.

```
(c:ace_multipole "HDS11" 3 (list 5.5 6.6 0.0) 1)
```

Example 3

Insert 3-pole disconnect switch at 5.5,6.6, force "vertical" version of symbol and build the switch to the right.

```
(c:ace_multipole "HDS11" 3 (list 5.5 6.6 0.0) 21)
```

Example 4

Insert 4-pole device consisting of a N.O. pushbutton, a N.C. contact, a N.O. contact, and a red pilot light.

```
(c:ace_multipole (list "HPB11" "HPB22" "HPB21" "HLT1R") 4 (list 5.5 6.6 0.0) 1)
```

Example 5

Insert 2-pole circuit breaker into empty space at 5.5,6.6 but temporarily override the drawing's default ladder rung spacing. Insert poles 3.0 units apart.

```
(setq GBL_wd_m (wd_1_nth_subst 11 "3.0" GBL_wd_m))
(c:ace_multipole "HCB1" 2 (list 5.5 6.6 0.0) 1)
(c:wd_reread_dwg_params)
```

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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_pnl2sch_lookup

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Description

Find schematic component symbol name. For a given panel footprint symbol, attempt to find the equivalent schematic component symbol name. This data is returned by a query on the schematic_lookup.mdb file.

AutoLISP

```
(c:ace_pnl2sch_lookup ben)
```

Parameters

Parameters	Description
ben	a panel footprint block entity name on the active dra

Returns

List of sublists with 2 elements each ((symbol_name comment) (symbol_name comment)) or nil if none found.

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c:ace_projwide_util_dbx

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Description

Project-wide update/retag function.

Parameters

Parameters	Description
dist	list of drawings from active project or nil=process all drawings in project
what2dolst	A list of up to 27 elements: nth 0 - run_retag = 1 for component retag, 0 to skip nth 1 - run_wire = 1 for wire renumbering, 0 to skip nth 2 - run_xref = 1 for cross-reference update, 0 to skip nth 3 - future, set to 0 nth 4 - run_ladr = 1 ladder reference resequence, 0 to skip nth 5 - run_sht = 1 sheet resequence, 0 to skip nth 6 - ladr_mode = "S" for sequential, "B" for bump up/down amount nth 7 - ladr_amt = for "B" above - bump sequencing down (integer number) nth 8 - ladr_1st = for "S" above - first ladder number (character string value) nth 9 - ladr_skip = if ladr_mode is "S" amount to skip an integer number nth 10- sht_mode = "S" for sequential renumbering, to bump current up/down nth 11- sht_1st = if sht_mode is "S" first sheet number nth 12- sht_amt = if sht_mode is "B" amount to bump or down as an integer nth 13- ladr_seq_mode = if ladr_mode is "S", 0 for continuous, 1 to skip nth 14- run_%D = 1 to adjust drawing properties Drawing %D value, 0 to skip nth 15- drawing%D = value for above nth 16- comp_tagmode = drawing properties component retag mode ("S" for sequential, "R" for reference-based) nth 17- comp_tagformat = drawing properties component tag format for above nth 18- comp_tag_seqstart = drawing properties sequential mode starting value (character string value) nth 19- wnum_mode = drawing properties wire numbering mode ("S" for sequential, "R" for reference-based) nth 20- wnum_format = drawing properties wire numbering format (ex: "%N") nth 21- wnum_seqstart = drawing properties sequential mode starting value (character string value) nth 22- wnum_seq_incr = drawing properties sequential mode increment value (integer) nth 23- cross_ref_fmt_upd = 1 to update drawing properties cross-ref format, 0 to skip nth 24- cross_ref_fmt = value for above nth 25- inter_cross_ref_fmt_upd = 1 to update inter drawing cross-ref format, 0 to skip nth 26- inter_cross_ref_fmt = value for above
data1st	(list wire_def_1st tb_def_1st) where tb_def_1st is reserved for future use, and where wire_def_1st = list of 8 elements for wire numbering options (set to nil if wire number not selected). nth 0 - overwrite = 0 for tag only new tag all nth 1 - wire_mode = "S" for sequential, "C" for sequential consecutive, "B" for reference-based

nth 5 - w_an_fmt_fixed = 1 to insert wire numbers as
fixed, nil for normal
nth 6 - dosigref = 1 cross-ref signals, 0 to skip
nth 7 - freshen = 1 to freshen database for signals, 0
skip

options_lst	For future use.
-------------	-----------------

Example 1

Run Component retag and Cross-reference update on whole project.

```
(c:ace_projwide_util_dbx nil (list 1 0 1) (list nil nil) nil)
```

Example 2

Adjust cross-reference settings on every drawing in active project.

```
(c:ace_projwide_util_dbx nil (list 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 "%.N" 1 "%S.  
(list nil nil)  
nil)
```

Example 3

Run Wire numbering project-wide, force overwrite and force reference-base tags.

```
(c:ace_projwide_util_dbx nil (list 0 1) (list (list 1 "R") nil) nil)
```

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Description

Re-tag schematic components on all drawings in "dlist" or in entire active project (if "dlist" is nil). This is done in a non-scripting mode.

Parameters

Parameters	Description
dist	nil for all dwgs in active project, otherwise = list of f path/dwg names to process. Use double backslashes single forward slashes.
options_lst	For future use.

Example

Retag components on all dwgs of active project.

```
(c:ace_retag_project nil nil)
```

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AutoCAD Electrical 2025: AutoLISP Reference
c:wd_calc_comptag
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Description

Calculate tag-ID for parent schematic component.

AutoLISP

```
(c:wd_calc_comptag param_lst)
```

Parameters

Parameters	Description
param_lst	(list "ben") where "ben" = parent component entity name.

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AutoCAD Electrical 2025: AutoLISP Reference
c:wd_delsym_main
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Description

Erase connected component "ben" and try to reconnect all broken wires and reconcile duplicate wire numbers.

AutoLISP

```
(c:wd_delsym_main ben)
```

Parameters

Parameters	Description
ben	AutoCAD entity name of selected AcadE component.

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c:wd_get_pinlist
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Description

```
(c:wd_get_pinlist mfg cat assycode)
```

Parameters

Parameters	Description
mfg	Manufacturer code (optional).
cat	Catalog part number (required).
assycode	assycode value (optional)

Returns

nil if no match found.

Example 1

To return the list: ("K1,K2" "0,A1X,A1Y;0,A4X,A4Y" "" "") :

```
(c:wd_get_pinlist "AB" "700-P200A48" nil)
```

Example 2

This small utility checks selected symbol block inserts and updates TEXTVALS and parent symbol PINLIST data as appropriate.

```
(defun c:textvals_upd ( / ss ix slen en x textvals textvals_combo_lst
  textvals_cache mfg cat asmb)
  (princ "\nSelect AcadE objects for TEXTVALS and PINLIST update")
  (setq ss (ssget '((0 . "INSERT"))))
  (if (/= ss nil)
    (progn
      (setq slen (sslenght ss))
      (setq ix 0)
      (while (< ix slen)
        (setq en (ssname ss ix))
        ; Check if component carries non-blank CAT/MFG attribute values.
        ; If so, do a query on catalog lookup and retrieve any TEXTVALS
        ; that might be found for the matching catalog lookup record.
        (setq cat (c:wd_getattrval en "CAT"))
        (if (AND cat (/= cat ""))
          (progn
            (setq mfg (c:wd_getattrval en "MFG"))
            (if (AND mfg (/= mfg ""))
              (progn
                (if (not (setq asmb (c:wd_getattrval en "ASSYCODE")))
                  (setq asmb "") ; no ASSYCODE attrib, set to blank
                )
              (setq textvals nil)
              ; Check temporary "cache"
              (if (setq x (member (list mfg cat asmb) textvals_combo_lst))
                (progn
                  (setq textvals (nth (- (length textvals_combo_lst) (length x))
                    textvals_cache))
                )
              )
            ; ELSE
```

```

    )
  )
  (if textvals
    (progn
      (foreach x textvals
        ; return list is (list (list attrnam attrval) (list attrnam attrval)...)
        (if (AND (car x) (cadr x) (/= (car x) "") (/= (cadr x) ""))
          (progn
            (if (c:wd_modattrval en (car x) (cadr x) nil)
              (progn
                (princ "nTEXTVALS ")
                (princ (car x))
                (princ " --> ")
                (princ (cadr x))
              )
            )
          )
        )
      )
    )
  )
  ; Look for PINLIST info. Rtrn= (list "coilpin1,coilpin2"
  ; "type1,pin1a,pin1b;type2,pin2a,pin2b;...")
  (if (setq x (c:wd_get_pinlist mfg cat asmb))
    (progn
      (if (AND (car x) (/= (car x) "")) ; coil pins, comma delimited
        (progn
          (if (setq xxx (c:wd_delim_str_to_lst (car x) ","))
            (progn ; now write coil pins out to TERM01, TERM02...
              (setq xx 1)
              (foreach pinval xxx
                (cond
                  ((< xx 10)
                   (c:wd_modattrval en (strcat "TERM0" (itoa xx)) pinval nil)
                  )
                  (T (c:wd_modattrval en (strcat "TERM" (itoa xx)) pinval nil))
                )
              )
              (setq xx (1+ xx))
            )
          )
        )
      )
    )
  )
  ; Put child pin info on component's PINLIST attrib or as xdata
  (if (AND (cadr x) (/= (cadr x) ""))
    (progn
      (if (c:wd_upd_pnlval en "PINLIST" (cadr x))
        (progn
          (princ "nPINLIST --< ")
          (princ (cadr x))
        )
      )
    )
  )
  )
  )
  )
  (setq ix (1+ ix))
)
(setq ss nil)
))
(setq textvals_combo_lst nil)
(setq textvals_cache nil)
(princ)
)

```

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Description

User hook into the "INSERT CIRC" command to insert a collection of prewired components. AcadE automatically ret inserted components based upon circuit line references or next available sequentials.

AutoLISP

```
(c:wd_ins_circ2 dwgnam xy scl flgbits)
```

Parameters

Parameters	Description
dwgnam	WBlocked circ to insert (i.e. AutoCAD ".dwg" file). If full path is not given then the normal AcadE and Aca paths are searched.
xy	Insertion coordinate (nil=user prompted)
scl	Insertion scale (default = 1.0)
flgbits	1's bit set = preserve "FIXED" wire numbers. 2's bit set = keep any SRC arrow symbols. 4's bit set = suppress check of component attribute assignments. 8's bit set = suppress check for symbol reconnection existing wires. 16's bit set = make sure that all LINE entities are m to valid wire layer. 32's bit set = don't blank out tags on orphan contact come in with the inserted circuit. 64's bit set = save copy of parent's TAG1 non-tagge default value as xdata value WD_VIA_PRETAG. This as temporary storage for the Circuit Builder utility. 128's bit set= adjust insertion point to lie on underly LINE entity

Returns

No return value.

Example 1

Insert 3-phase motor circuit "FVR_KFDS.dwg" at XY 15,18.5 with scale 1.0, preserve SRC arrows and make sure all on valid wire layers.

```
(c:wd_ins_circ2 "FVR_KFDS" (list 15.0 18.5) 1.0 18)
```

Example 2

This snippet of code wraps around the AcadE INS CIRC command. It prompts for the "suffix" to be applied to each terminal tag and device tag in the inserted circuit. The key is that the terminal and component tags need to be end


```

circ_ents_added_list)
(setq circ_to_insert nil)
(setq circ_insertion_xy nil)
(setq options 33) ; 1's bit set=keep fixed wire nums, 32's set=don't blank orphans
(setq suffix (getstring " SUFFIX value="))
(if (AND suffix (/= suffix ""))
    (progn
      (setq #lastent (c:ace_lastent_mark)) ; remember last entity in database
      ; Insert the circuit
      (c:wd_ins_circ2 circ_to_insert circ_insertion_xy 1.0 options)
      ; Now process the block inserts that went in with the circuit
      (setq circ_ents_added_list (c:ace_lastent_list_from #lastent 0)) ; get INSERTS added
      ; "circ_ents_added_list" = list of INSERTS that went in with the circuit
      (foreach en circ_ents_added_list
        ; Look for TAG1, TAG2, TAGSTRIP attribute
        (setq val (c:wd_getattrval en "TAG1,TAG1F,TAG2"))
        (if (AND val (/= val ""))
            (progn
              ; Check last character for target flag of "%"
              (if (= (substr val (strlen val) 1) "%")
                  (progn ; Yes, make the substitution
                    (setq newval (strcat (substr val 1 (1- (strlen val))) suffix))
                    (c:wd_modattrval en "TAG1,TAG1F,TAG2" newval nil)
                  ))
            ))
        (if (c:wd_getattrval en "TAGSTRIP")
            (progn ; hit a terminal
              (setq val (c:wd_getattrval en "TERM*")) ; get terminal number
              ; Check last character for target flag of "%"
              (if (= (substr val (strlen val) 1) "%")
                  (progn ; Yes, make the substitution
                    (setq newval (strcat (substr val 1 (1- (strlen val))) suffix))
                    (c:wd_modattrval en "TERM*" newval nil)
                  ))
            ))
        ))
      ))
    (princ)
  )

```

Module

[Section A - Schematic Components](#)

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insert_elect_block

[Example 1](#) [Example 2](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

Generic INSERT BLOCK command. Wires do NOT break under the block.

AutoLISP

Parameters	Description
sym	Symbol block name (AcadE's search path is used wh path name is omitted)
xy	Insertion coordinate - not optional.
layflg	nil to keep attribs on existing layer, otherwise move attributes to layers per drawing's CONFIG setting (W block).
scl	Insertion scale (default = 1.0)

Returns

nil or entity name of successfully inserted block insert.

Example 1

```
(c:wd_insert_elect_block "hcr1" (list 5 6) 1 1.0)
```

Example 2

```
(c:wd_insert_elect_block "hcr1" (getpoint) 1 1.0)
```

Module

[Section A - Schematic Components](#)

Links

[Example 1](#), [Example 2](#), [Section A - Schematic Components](#)

Description

User hook into the INS COMPONENT command, command-line prompted. Same as [c:wd_insym2](#) but command pror for symbol name typed in at command line and an insertion point. All data is supplied on the command line with op EDIT dialog box popping up after symbol inserts. Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically generates based on inserted lin next available sequential.

Author

Returns

ent name or "handle" (see "options") of inserted symbol or nil if fail.

Module

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insym_dlg

[Example 1](#) [Example 2](#) [Example 3](#) [Example 4](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

User hook into the INS COMPONENT command, icon menu driven. Same as wd_insym but command triggers icon n display for component selection, user selects insertion point. The icon menu that pops up is the schematic icon mer is in memory or is the default schematic icon menu (if the menu isn't already in memory. Component automatically any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatical generates based on inserted line ref or next available sequential.

AutoLISP

```
(c:wd_insym_dlg)
```

Returns

No return value.

Example 1

Pops open the current/default schematic icon menu.

```
(c:wd_insym_dlg)
```

Example 2

Pops open a specific icon menu one time.

```
(c:wd_loadmenu "c:/my_custom/my_special_menu.dat")  
(c:wd_insym_dlg)
```

```
(setq GBL_wd_sch_menu_override "c:/my_custom/my_special_menu.dat")
```

Example 4

Flip back to the default schematic icon menu.

```
(setq GBL_wd_sch_menu_override nil)
```

Module

[Section A - Schematic Components](#)

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[Example 1](#), [Example 2](#), [Example 3](#), [Example 4](#), [Section A - Schematic Components](#)

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insym_f

[Section A - Schematic Components](#) [Send Feedback](#)

Description

User hook into the INS COMPONENT command, prompt to select "insert fence" component. User prompted to select fence" component from icon menu. Then user prompted to draw fence. AcadE inserts selected component at interse between fence and underlying wires.

AutoLISP

```
(c:wd_insym_f)
```

Returns

Ent name of inserted symbol or of deleted symbol if cancel out of the EDIT dialog box. Returns nil if symbol not four (AcadE searches in its normal list of paths for the symbol).

Module

[Section A - Schematic Components](#)

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Description

User hook into the INS COMPONENT command, user prompted to pick on a "just like" component. Same as wd_insy but user prompted to pick on a "just like" component. Then user prompted to draw fence. AcadE inserts selected component at intersections between fence and underlying wires.

AutoLISP

```
(c:wd_insym_f_same)
```

Returns

Ent name of inserted symbol or of deleted symbol if cancel out of the EDIT dialog box. Returns nil if symbol not fou (AcadE searches in its normal list of paths for the symbol).

Module

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insym_go2menu
[Example](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

Display a submenu page in the current icon menu. Jump to and display the "mnum" submenu page in the current ic menu file. A pick on this menu will feed the selected button's value to the INS COMPONENT command.

AutoLISP

```
(c:wd_insym_go2menu mnum)
```

Parameters

Parameters	Description
mnum	Menu page number to display (i.e. the "***Mx" numb the wd_menu.dat file, or whatever menu is current).

Module

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[Example](#), [Section A - Schematic Components](#)

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insym_gomenu

[Example](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

Jump to and display the "mnum" submenu page in the current icon menu file. A pick on this menu will feed the sele button's value to the INS COMPONENT command.

AutoLISP

```
(c:wd_insym_gomenu mnum)
```

Parameters

Parameters	Description
mnum	Argument mnum is passed seperately. Menu page n to display (i.e. the "**Mx" numbers in the wd_menu file, or whatever menu is current).

Example

See the acad.e.cui file.

Module

[Section A - Schematic Components](#)

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[Example](#), [Section A - Schematic Components](#)

Description

User hook into the INS COMPONENT command, just-like component, insert prompted. Same as wd_insym but user prompted to select a "just like" component (schematic or panel footprint). Then user is prompted for insertion point INS/EDIT dialog box appears after insert. The icon menu that pops up is the schematic icon menu that is in memory the default schematic icon menu (if the menu isn't already in memory. Component automatically breaks any underlying wire(s), new wire number generates on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.

AutoLISP

```
(c:wd_insym_same)
```

Returns

Ent name of inserted symbol or of deleted symbol if cancel out of the EDIT dialog box. Returns nil if symbol not found (AcadE searches in its normal list of paths for the symbol).

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insym2

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Description

User hook into the INS COMPONENT command. All data is supplied on the command line with optional EDIT dialog box popping up after symbol inserts. Component automatically breaks any underlying wire(s), new wire number generates new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.

AutoLISP

```
(c:wd_insym2 sym xy scl options)
```

Parameters

Parameters	Description
------------	-------------

xy	xy insertion point coordinate, nil=user prompted for insertion point.
scl	scale (nil=1.0)
options	1's bit set = pop open "INSERT/EDIT" dialog box after insertion. 2's bit set = return "handle" of inserted symbol (otherwise return entity name). 4's bit set = (and when 1's bit = 0) suppress parent component or terminal tag auto-generation. 8's bit set = allow motor angled connections to go to wire LINE entity (used by CIRC BLDR).

Returns

entity name or "handle" (see "options") of inserted symbol or nil if fail.

Example

To insert symbol "HPB11" into a wire at XY loc 5,6.25 at default scale and suppress popping up the INS/EDIT dialog afterwards.

```
(setq new_entname (c:wd_insym2 "HPB11" (list 5.0 6.25) nil nil))
```

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[Example](#), [Section A - Schematic Components](#)

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_insymn

[Example](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

User hook into the INS COMPONENT command, name via command-line, insert prompted. Same as wd_insym but component name supplied on command line, user prompted to select insertion point. INS/EDIT dialog box displays insert.The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon menu if the menu isn't already in memory. Component automatically breaks any underlying wire(s), new wire number generated on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.

AutoLISP

Parameters	Description
sym	Symbol block name (AcadE's search path is used wh path name is omitted). Block name is auto-adjusted, possible, to align with an underlying wire.

Returns

Ent name of inserted symbol or of deleted symbol if cancel out of the EDIT dialog box. Returns nil if symbol not fou (AcadE searches in its normal list of paths for the symbol).

Example

```
(c:wd_insymn "HPB11")
```

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AutoCAD Electrical 2025: AutoLISP Reference
c:wd_insymn_repeat
[Section A - Schematic Components](#) [Send Feedback](#)

Description

User hook into the INS COMPONENT command, name via command-line, insert prompted, insertion of sym repeate Keeps repeating the insert of selected "sym". Component name supplied on command line, user prompted to select insertion point and will attempt to auto-align with any underlying wire. The Insert/Edit dialog box appears after eac insert. The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon m the menu isn't already in memory. Component automatically breaks any underlying wire(s), new wire number gene on new wire segment (if required), component tag automatically generates based on inserted line ref or next availa sequential.

AutoLISP

```
(c:wd_insymn_repeat sym)
```

Parameters

Parameters	Description
------------	-------------

Returns

nil.

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AutoCAD Electrical 2025: AutoLISP Reference
c:wd_insymn_rot
[Example 1](#) [Example 2](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

User hook into the INS COMPONENT command, name via command-line, insert prompted, rotation prompted. Component name supplied on command line, user prompted to select insertion point, but if the device cannot find a wire/line connection to tie to, it then prompts the user for an insertion ROTATION value. The INS/EDIT dialog box opens after insert. The icon menu that pops up is the schematic icon menu that is in memory or is the default schematic icon menu if the menu isn't already in memory. Component automatically breaks any underlying wire(s), new wire number generated on new wire segment (if required), component tag automatically generates based on inserted line ref or next available sequential.

AutoLISP

```
(c:wd_insymn_rot sym)
```

Parameters

Parameters	Description
sym	Symbol block name (AcadE's search path is used when path name is omitted). Block name is auto-adjusted, possible, to align with an underlying wire.

Returns

Entity name of inserted symbol or of deleted symbol if cancel out of the EDIT dialog box. Returns nil if symbol not found.

Example 1

Here we open the current/default schematic icon menu:

Pops open a specific icon menu.

```
(c:wd_loadmenu "c:/my_custom/my_special_pneumatic.dat")
(c:wd_insymn_rot "HSV1_PNEU")
```

Module

[Section A - Schematic Components](#)

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[Example 1](#), [Example 2](#), [Section A - Schematic Components](#)

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_loadmenu
[Example 1](#) [Example 2](#) [Example 3](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

Activate AcadE icon menu. Make AcadE icon menu file "fnam" the active icon menu (for INSERT COMPONENT using c:wd_insym_dlg, c:wd_insym_f, and others).

AutoLISP

```
(c:wd_loadmenu fnam)
```

Parameters

Parameters	Description
fnam	Icon menu ascii data file (= nil defaults to "wd_menu.dat").

Example 1

Pop open a specific schematic icon menu (one time):

```
(c:wd_loadmenu "c:/my_custom/my_special_menu.dat")
(c:wd_insym_dlg)
```

Example 2

Make a specific schematic icon menu persist during the current AcadE session:

```
(setd GBT wd sch menu override "c:/my_custom/my_special_menu.dat")
```

```
(setq GBL_wd_sch_menu_override nil)
```

Module

[Section A - Schematic Components](#)

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_menu_load_schem

[Example](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

Loads the default schematic icon menu .dat file is loaded and "in memory" so that it will display for a subsequent c: related to INS COMPONENT. The menu that is loaded is determined in this order: 1st - an override menu name is defined in global GBL_wd_sch_menu_override. 2nd - defined for the active project (right-click on project's "Symbol Libraries" 3rd - defaults to menu file "ace_jic_menu.dat". This routine can be used in ".cui" file calls to "wd_insym_gomenu" so that AcadE will display schematic icon menu pages and not menu pages from the last icon menu file used (if it was a schematic icon menu file).

AutoLISP

```
(c:wd_menu_load_schem)
```

Example

See the ace_only.mnu or ace2006.mnu for examples.

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_movesym2

[Example](#) [Section A - Schematic Components](#) [Send Feedback](#)

aligns with and breaks underlying wire(s), new wire number generates on new wire segment (if required), compone can automatically generate based on inserted line ref or next available sequential. Wire "heals" where component n from.

AutoLISP

```
(c:wd_movesym2 ben xy options)
```

Parameters

Parameters	Description
ben	Ent name of block insert to move. nil = user prompt
xy	xy "move to" insertion point coordinate (nil=user prompted)
options	1's bit set = suppress "re-tag" check. 2's bit set = return "handle" of inserted symbol. (not set = ent name) nil = open Component(s) Moved dialog box promptir user for retag or no retag

Returns

Handle of newly moved component

Example

To move the last inserted symbol to at XY loc 10,10.
(c:wd_movesym2 (entlast) "10,10", 2)

Module

[Section A - Schematic Components](#)

Links

[Example](#), [Section A - Schematic Components](#)

Description

```
(c:wd_pinlist_attach ben options)
```

Parameters

Parameters	Description
ben	Ent name of schematic block insert to process.
options	1's bit set = update "coil" pin (TERMxx) attributes or symbol.

Module

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AutoCAD Electrical 2025: AutoLISP Reference
c:wd_retag_update_noprompts
[Section A - Schematic Components](#) [Send Feedback](#)

Description

Re-tag schematic component(s) on current dwg only and optionally updates related child components across active project.

AutoLISP

```
(c:wd_retag_update_noprompts options ent_lst)
```

Parameters

Parameters	Description
options	0 = user prompt to select components to retag, user prompted for OK to check other dwgs in project for r children (unless bit 32 set). 1's bit to do dwg-wide retag update without prompt 2's also do project wide retag update without prompt 4's bit set = DESCx was changed on parent and not else. 8's bit set = DESCx changed on parent along with T/ CAT, or something else significant. 16's bit set = processing/updating CHH D cross-refs

ent_lst	(with 1's bit set) nil = do all components on dwg, OF do ents passed in this argument. Can be single entity name, entity handle, list of entity names, list of entity handles, or selection set.
---------	--

Remarks

Replaces obsolete functions wd_update_circ_noprmts and wd_retag_comp.

Module

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AutoCAD Electrical 2025: AutoLISP Reference
c:wd_xref_doit
[Example](#) [Section A - Schematic Components](#) [Send Feedback](#)

Description

Run project-wide parent/child cross-reference text update. Note: the drawing or drawings to be processed must be the active project.

AutoLISP

```
(c:wd_xref_doit dlst child_update no_report_dlg)
```

Parameters

Parameters	Description
dlst	List of dwgs to process (full file names, double back slashes) nil = process all dwgs in current project set
child_update	1's bit set to do DESC update. 2's bit set to do just blank child DESC update. 4's bit set to do child LOC update.
no_report_dlg	1 to suppress cross-ref exception report dlg. nil = show report

```
(setq dlst (nth 5 (c:wd_proj_wdp_data)))  
(command "_QSAVE") ; to suppress any prompt for "save changes"  
(c:wd_xref_doit dlst 2 1) ; update blank child DESC
```

Module

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